```
4018b5:    f3 48 ab       rep stos %rax,%es:(%rdi)
401901:    66 0f ae 3a    clflushopt (%rdx)
401913:    66 0f ae 38    clflushopt (%rax)
401930:    66 0f ae 38    clflushopt (%rax)
401938:    66 0f ae 3a    clflushopt (%rdx)
401945:    0f ae f0       mfence
```

TID — The Instant Destroyer | Functional Test Report

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Generated : 2026-05-16 22:15:07 UTC
CPU : model name : AMD EPYC 9B14
Kernel : 6.14.11
Compiler : gcc (GCC) 14.3.0
CLFLUSHOPT : PRESENT ✓
Build Flags: gcc -O2 -mclflushopt

HOW TID ZEROIZES MEMORY — 3 PHASES

```
Phase 1 — REP STOSQ     : writes zeros 8 bytes per cycle
                          compiler cannot remove this (inline asm)
Phase 2 — CLFLUSHOPT    : evicts every 64-byte cache line
                          removes data from L1 / L2 / L3
Phase 3 — MFENCE        : full memory barrier
                          ensures all evictions complete before return
```

DISASSEMBLY PROOF — key instructions present in -O2 binary

FUNCTIONAL TEST SUITE — correctness on 29 buffer sizes

Pattern: buffer filled with 0xAB, then zeroized, then verified

```
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 [PASS] size=1           370 ns  all bytes = 0x00 ✓
 [PASS] size=3           240 ns  all bytes = 0x00 ✓
 [PASS] size=7           250 ns  all bytes = 0x00 ✓
 [PASS] size=8           250 ns  all bytes = 0x00 ✓
 [PASS] size=9           250 ns  all bytes = 0x00 ✓
 [PASS] size=15          240 ns  all bytes = 0x00 ✓
 [PASS] size=16          240 ns  all bytes = 0x00 ✓
 [PASS] size=17          240 ns  all bytes = 0x00 ✓
 [PASS] size=31          260 ns  all bytes = 0x00 ✓
 [PASS] size=32          250 ns  all bytes = 0x00 ✓
 [PASS] size=33          230 ns  all bytes = 0x00 ✓
 [PASS] size=37          250 ns  all bytes = 0x00 ✓
 [PASS] size=63          240 ns  all bytes = 0x00 ✓
 [PASS] size=64          240 ns  all bytes = 0x00 ✓
 [PASS] size=65          260 ns  all bytes = 0x00 ✓
 [PASS] size=100         260 ns  all bytes = 0x00 ✓
 [PASS] size=127         250 ns  all bytes = 0x00 ✓
 [PASS] size=128         250 ns  all bytes = 0x00 ✓
 [PASS] size=256         260 ns  all bytes = 0x00 ✓
 [PASS] size=511         280 ns  all bytes = 0x00 ✓
 [PASS] size=512         270 ns  all bytes = 0x00 ✓
 [PASS] size=1024        280 ns  all bytes = 0x00 ✓
 [PASS] size=4095        430 ns  all bytes = 0x00 ✓
 [PASS] size=4096        420 ns  all bytes = 0x00 ✓
 [PASS] size=65535       3340 ns  all bytes = 0x00 ✓
 [PASS] size=65536       3330 ns  all bytes = 0x00 ✓
 [PASS] size=131072      8300 ns  all bytes = 0x00 ✓
 [PASS] size=524288      31630 ns  all bytes = 0x00 ✓
 [PASS] size=1048576     54670 ns  all bytes = 0x00 ✓
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Results: 29 passed, 0 failed
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PERFORMANCE BENCHMARK — 1000 iterations per size, warm-up included
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 Size      | min ns      | avg ns      | max ns      | GB/s (avg)
 -----------|-------------|-------------|-------------|-------------
```

32 B	240	372	43650	0.09
64 B	280	307	630	0.21
256 B	280	346	630	0.74
1 KB	260	314	14560	3.26
4 KB	400	503	10930	8.14
16 KB	960	1081	1400	15.16
64 KB	3170	3731	16450	17.57
256 KB	12300	14345	121310	18.27
1 MB	45510	54875	145740	19.11

CONCLUSION

- All 29 sizes zeroized correctly (0 failures)
- REP STOSQ + CLFLUSHOPT + MFENCE survive -O2 optimization
- Peak throughput: ~20 GB/s on 256 KB – 1 MB buffers
- Sub-microsecond latency for buffers below 4 KB